Table of Contents

Abstract 2
Introduction and Mathematical Modelling 2
Methodology 5
Results and Discussion 9
Research Question 1 9
Research Question 2 13
Research Question 3 17
Limitations 19
Conclusion 19
Reference 20
Appendix 22

Abstract

The aim of this report is to create a model which depicts the propagation of pollution in the Great Lakes located in east-central North America.[1] The model is able to illustrate the change in concentration of pollutants in each lake over time, more specifically, between 1950 and 2019. The Great Lakes comprise of Lake Superior, Lake Michigan-Huron, Lake Erie, and Lake Ontario. Since the start of the 1950s the lakes were heavily polluted by industries and by water runoff which picked up fertilizers from nearby agricultural lands. The model was created by using data recorded every month for 70 years since 1950. This model would allow us to visualize which lake is more vulnerable to pollution and therefore be able to provide treatment to the affected lakes or apply quotas to industries.

Firstly, we will utilise the model created to simulate the downstream propagation of pollution in the Great Lakes for three scenarios. The first scenario involves Lake Superior being polluted with 75,000 metric tons of pollution in 1950 whilst the other three lakes were not polluted and the concentration of pollutants in water runoff is assumed to be zero. The second and third scenario are similar to the first, however, in the second scenario Lake Superior is polluted with 50,000 metric tons of pollution in 1950. In the last scenario Lake Superior is polluted with 25,000 metric tons of pollution in 1950. We will then analyse and draw conclusions to the results shown in the model.

The second problem involves evaluating the effect of water inflow, outflow, and runoff in the concentration of pollution in the water of each lake. It requires us to identify which of the three factors, which are important in the mechanism of pollution, is the defining factor in the concentration of pollution. Additionally, we will lift the assumption that the concentration of pollutants in water runoff is zero by approximating its value through data given of phosphorus levels in different types of farms. The initial mass of pollutants will be assumed to be zero, so we can focus solely on the effects of polluted water runoff.

Lastly, the model was very much focused on investigating the propagation of pollution and the approximate effects of phosphorus in the lakes. However, many other factors may affect the lakes instead of nutrient pollution. It is required to find the concentration of pollutants (phosphorus) in the water runoff which would lead to a eutrophic state of the lakes (concentrations of phosphorus above 20 μg per litre), meaning an environment which is favorable for algae blooms. In the second problem we approximated the concentration of phosphorus in water runoff and found which of the three factors are the defining factor on the concentration of pollution in the lakes. By calculating the harmful concentration of phosphorus in the lakes in this problem we can conclude whether phosphorus is the main problem in lake pollution. This is done by comparing the two concentrations found in the second and last problem. We will then discuss whether any imposition of limitations to the recommended threshold on the concentration of phosphorus per kilogram of soil would provide with any type of aid.

Introduction and Mathematical Modelling

Humans have always strived to improve efficiency and capability. For example, farmers today are able to increase their yield efficiency and total yield through large-scale farming, and better machinery but also with the use of fertilizers to improve the growth of plants. [2] However, such inventions may lead to other problems in the

world. The use of fertilizers such as phosphorus, to improve root development and fruit development and nitrogen may lead to eutrophication which is a result of nutrients used in agriculture being transported through under/over groundwater run-off into lakes. Nitrogen is abundant in nature; however, phosphorus is not thus a high concentration of this nutrient in a natural environment may cause serious problems. Eutrophication is the rapid and large growth of undesirable algae in lakes. Algae is necessary for the food web; however, large algae growth may lead to serious issues. Thick layers of algae rise to the surface and block any incoming light into the lakes. This kills the plants in the lakes which release oxygen. Additionally, the algae decompose and produce large amounts of carbon dioxide. [3] The lack of oxygen and overproduction of carbon dioxide leads to lake acidification thus killing fish and other aquatic organisms. This raises serious problems in the Great Lakes because the Great lakes hold approximately 21% of the planet's freshwater by volume and is essential for our water supply and power generation. The Great Lakes are essential for industries, agriculture, and food production and are home to significant aquatic biodiversity. The death of such lakes would raise serious economic, developmental, and food supply. Since the 1990s one of the lakes, Lake Erie, has experienced large algae growth which resulted in poor water quality and impairment of natural resources [4] such as food and water supply for industries. Actions need to be taken to minimise the large growth of algae as they impact wildlife and our industries.

Our aim is to create a model to display the concentration of pollutants in the lakes over time and the total mass of pollutants, with recorded data from 1950 up until 2019. This sort of model may be useful to experts who are involved in water quality.

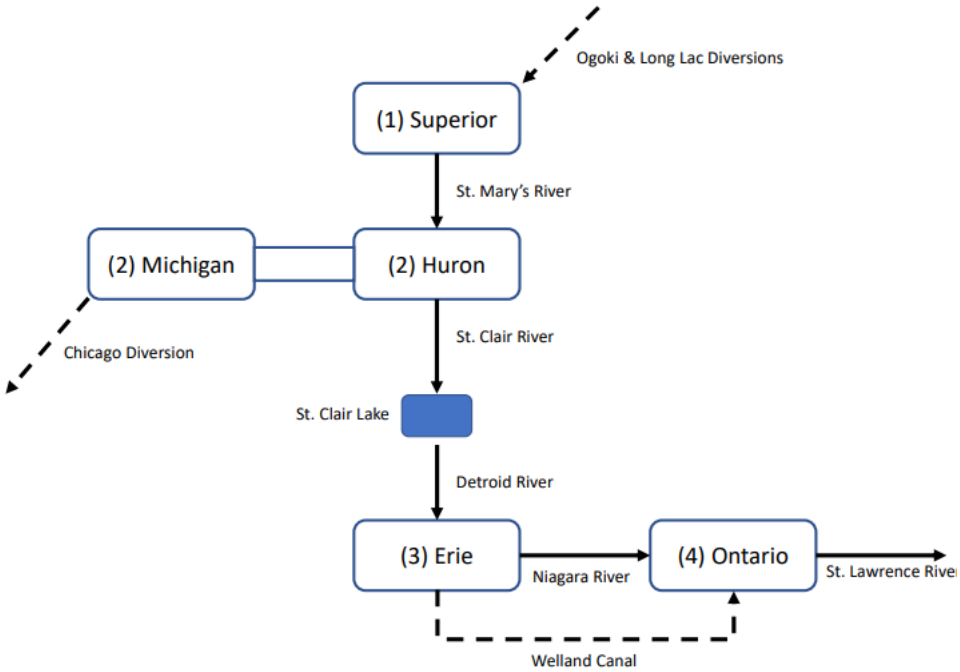

Figure 1: Great Lakes network

In order to create such a model, the volume of water entering and leaving each lake must be measured and recorded. From 1950 to 2019 the volume of water entering and leaving the lakes was recorded every month for 70 years. The inflow of water comes from precipitation falling directly into the lake (P), groundwater runoff (R),

connecting the flow of water from upstream lakes (previous lake to current lake) (F^{in}) and any existing diversions into the lake (D^{in}). The outflows of the lakes come from, evaporation from the surface of the lake (E), connecting the flow of water into downstream lakes (current lake to next downstream lake) (F^{out}) and any existing diversions out of the lake (D^{out}).

All of these parameters change over time as they depend on many factors such as seasons, therefore, the change in volume is equal to:

$$\frac{dV_1}{dt} = P_1 + R_1 + D_1^{in} - E_1 - F_1^{out} \tag{1}$$

$$\frac{dV_i}{dt} = P_i + R_i + F_{i-1}^{out} - E_i - F_i^{out} - D_i^{out},\ i = \{2, 3\} \tag{2}$$

$$\frac{dV_4}{dt} = P_4 + R_4 + F_3^{out} + D_3^{out} - E_4 - F_4^{out} \tag{3}$$

Where $i =$ the numbered Great Lakes in Fig. (1).

By using MATLAB, the previous year's Mathematics groups were able to calculate the total volume of water in each lake for each month. This was done by collecting all of the data for inflows and outflows to calculate the change in volume for each month. Then the total volume for each month was calculated by summing up all of the changes in the volume of water in the lake. We then used the calculations and recordings done by the previous year's group to find the total mass of pollutants in the lakes. The equation for the change in the mass of pollutants in the lakes is given:

$$\frac{dm_i}{dt} = c_{r,i}R_i + \frac{m_{i-1}}{V_{i-1}}T_{i-1}^{out} - \frac{m_i}{V_i}T_i^{out} \tag{4}$$

This equation is derived from the change in volume of water Eq. 2. The onflows may be expressed as:

$$T_{i-1}^{out} = P_i + R_i + F_{i-1}^{out}$$

where i-1 depicts the outflow of the upstream lake which has now become the inflow of the current lake. The outflows are depicted through the equation:

$$T_i^{out} = E_i + F_i^{out} + D_i^{out}$$

The concentration of pollutants is equal to the mass of pollutants per volume.

$$c = \frac{m}{V}$$

$$\text{So, } m = c * V$$

By differentiating both sides, we get

$$\frac{dm_i}{dt} = \frac{dc}{dt} * V + \frac{dV}{dt} * c = c * \frac{dV}{dt} = c * \left(T_{i-1}^{out} - T_i^{out}\right) = \frac{m_{i-1}}{V_{i-1}}T_{i-1}^{out} - \frac{m_i}{V_i}T_i^{out}$$

However, we have not included the change in mass of pollutants provided by the water runoff, therefore, by considering the concentration of pollutants in the water runoff and the change in volume of the water provided by water runoff (R)

$$\frac{dm_i}{dt} = c_{r,i}R_i + \frac{m_{i-1}}{V_{i-1}}T_{i-1}^{out} - \frac{m_i}{V_i}T_i^{out}$$

where m_i represents the total mass of pollutants, $c_{r,i}$represents the concentration of pollutants in the water runoff, and T_{i-1}^{out} represents the outflow from an upstream lake that carries pollution downstream.

In order to find the total mass of pollutants for each lake we used Euler's Method on MATLAB. We created a 'for' loop where it calculated the concentration of pollutants from 1950 until 2019. One month was our step length. The instantaneous slope function was Eq.(4) and the starting point was the mass of pollutants in the lakes at the start of 1950, this was assumed to be 0 kg. We calculated the change in the mass of pollutants by using the total volume data of the previous year's group. The concentration was approximated through the assistance of an article published by the University of Colorado. According to Euler's Method, to find the total mass we add the change in mass for each month to the mass of pollutants recorded in the previous month.

$$m_i(t+1) = m_i(t) + dm_i, \qquad i = \{1, 2, 3, 4\} \tag{5}$$

Where $m_i(t+1)$ is the mass of pollutants in the current month, $m_i(t)$ is the mass of pollutants in the previous month, dm_i is the change in mass of pollutants in the current month and i is the numbered lake, shown in Fig.(1).

This equation was derived through the approximation of the derivative:

$$\frac{dm_i}{dt} \approx \frac{m_i(t+1) - m_i(t)}{step\ length}, \qquad i = \{1, 2, 3, 4\} \tag{6}$$

$$\Rightarrow m_i(t+1) = m_i(t) + step\ length * dm_i \tag{7}$$

This loop will continue from 1st year (1950) up until the last year of recording (2019). Having calculated the total volume of water and the total mass of pollutants in each lake for each month we were able to calculate the approximate concentration of mass of pollutants in each lake by dividing the total mass of pollutants in a lake in a month by its corresponding volume at that month for all 840 months. The dumping of pollutants in the Great Lakes began in the 1900s, therefore, being able to create a model which simulates the movement of pollutants in the past we are able to observe which lakes were highly affected lakes and which were the least affected. Furthermore, if we are able to estimate the amount of pollutants that may get dumped in the lakes in the future, we can simulate the effects of them in the lakes. Therefore, by implementing quotas and laws for industries we can stop future effects from occurring, thus providing resilience and sustainability.

Methodology

Dimensional analysis:

Equation (4) in the modelling part is a general form of mass balance equation for all four lakes. Aiming to solve the equation for each lake, the first step is to find the bass balance equation for each lake:

Superior

$$\frac{dm_1}{dt} = c_{r,1}R_1 - \frac{m_1}{V_1}T_1^{out} \tag{8}$$

Michigan-Huron

$$\frac{dm_2}{dt} = c_{r,2}R_2 + \frac{m_1}{V_1}T_1^{out} - \frac{m_2}{V_2}T_2^{out} \tag{9}$$

Erie

$$\frac{dm_3}{dt} = c_{r,3}R_3 + \frac{m_2}{V_2}T_2^{out} - \frac{m_3}{V_3}T_3^{out} \tag{10}$$

Ontario

$$\frac{dm_4}{dt} = c_{r,4}R_4 + \frac{m_3}{V_3}T_3^{out} - \frac{m_4}{V_4}T_4^{out} \tag{11}$$

$T_1^{out}, T_2^{out}, T_3^{out}\ and\ T_4^{out}$, are given by the sum of an upstream lake's outflow from channels and diversions.

Therefore Equation (8) – Equation (11) can be rewritten as:

Superior

$$\frac{dm_1}{dt} = c_{r,1}R_1 - \frac{m_1}{V_1}F_1^{out} \tag{12}$$

Michigan-Huron

$$\frac{dm_2}{dt} = c_{r,2}R_2 + \frac{m_1}{V_1}F_1^{out} - \frac{m_2}{V_2}(F_2^{out} + D_2) \tag{13}$$

Erie

$$\frac{dm_3}{dt} = c_{r,3}R_3 + \frac{m_2}{V_2}F_2^{out} - \frac{m_3}{V_3}(F_3^{out} + D_3) \tag{14}$$

Ontario

$$\frac{dm_4}{dt} = c_{r,4}R_4 + \frac{m_3}{V_3}(F_3^{out} + D_3) - \frac{m_4}{V_4}F_4^{out} \tag{15}$$

the units of the given data outflow F_i^{out} $\{i = 1,2,3,4\}$, and divertion D_i are m^3/s. However the unit of water runoff R_i is in mm over the respective lake's surface area per month. In order to make sure the mass balance equations for each lake are dimensionally homogeneous, F_i^{out} and D_i and need to be converted to mm per unit area per month, which can be achieved by multiplying $F_i^{out}\ and\ D_i$ by:

$$\zeta_i = 60\left[\frac{s}{min}\right] \times 60\left[\frac{min}{h}\right] \times 24\left[\frac{h}{day}\right] \times 30\left[\frac{day}{month}\right] \times 1000\left[\frac{mm}{m}\right] \times \frac{1}{A_i}\left[\frac{1}{m^2}\right]$$

where A_i is the surface area of the respective lake.

This change can be made in Equations (12) – (15) so that they are all dimensionally homogeneous and in units of $kg/month$:

Superior

$$\frac{dm_1}{dt} = \left(c_{r,1}R_1 - \frac{m_1}{V_1}F_1^{out}\zeta_1\right)A_1 \times 10^{-3} \tag{16}$$

Michigan-Huron

$$\frac{dm_2}{dt} = \left(c_{r,2}R_2A_2 + \frac{m_1}{V_1}F_1^{out}\zeta_1A_1 - \frac{m_2}{V_2}(F_2^{out} + D_2)\zeta_2A_2\right) \times 10^{-3} \tag{17}$$

Erie

$$\frac{dm_3}{dt} = \left(c_{r,3}R_3A_3 + \frac{m_2}{V_2}F_2^{out}\zeta_2A_2 - \frac{m_3}{V_3}(F_3^{out} + D_3)\zeta_3A_3\right) \times 10^{-3} \tag{18}$$

Ontario

$$\frac{dm_4}{dt} = \left(c_{r,4}R_4A_4 + \frac{m_3}{V_3}(F_3^{out} + D_3)\zeta_3A_3 - \frac{m_4}{V_4}F_4^{out}\zeta_4A_4\right) \times 10^{-3} \tag{19}$$

Symbol	Lake	Surface Area [$\times 10^{10}$ m^2]	Initial Volume [km^3]
A_1	Superior	8.1925	12 000
A_2	Michigan-Huron	11.1685	8 400
A_3	Erie	2.5404	480
A_4	Ontario	1.9121	1 640

Table 1. Lake parameters.[7]

The parameter $c_{r,1}$:

When $\frac{dm_i}{dt}$ is equal to zero, it means there is no change in pollutant mass in that month, which also means the amount of pollutants that flow into a lake exactly equals the amount of pollutants flowing out of that lake. in this case, Equation (16) – (19) can be written as:

Superior

$$0 = \left(c_{r,1}R_1 - \frac{m_1}{V_1}F_1^{out}\zeta_1\right)A_1 \times 10^{-3} \tag{20}$$

Michigan-Huron

$$0 = \left(c_{r,2}R_2A_2 + \frac{m_1}{V_1}F_1^{out}\zeta_1A_1 - \frac{m_2}{V_2}(F_2^{out} + D_2)\zeta_2A_2\right) \times 10^{-3} \tag{21}$$

Erie

$$0 = \left(c_{r,3}R_3A_3 + \frac{m_2}{V_2}F_2^{out}\zeta_2A_2 - \frac{m_3}{V_3}(F_3^{out} + D_3)\zeta_3A_3\right) \times 10^{-3} \tag{22}$$

Ontario

$$0 = \left(c_{r,4}R_4A_4 + \frac{m_3}{V_3}(F_3^{out} + D_3)\zeta_3A_3 - \frac{m_4}{V_4}F_4^{out}\zeta_4A_4\right) \times 10^{-3} \tag{23}$$

The value of $c_{r,i}$, which contributes to the amount of pollutant flow into the lakes, can be obtained by rearranging Equation (20) – (23) with regards to the variable $c_{r,i}$ in each lake:

Superior

$$c_{r,1} = \frac{1}{R_1} \cdot \frac{m_1}{V_1}F_1^{out}\zeta_1 \tag{24}$$

Michigan-Huron

$$c_{r,2} = \frac{1}{R_2} \cdot \frac{m_2}{V_2}(F_2^{out} + D_2)\zeta_2 - \frac{1}{R_2} \cdot \frac{m_1}{V_1}F_1^{out}\zeta_1\frac{A_1}{A_2} \tag{25}$$

Erie

$$c_{r,3} = \frac{1}{R_3} \cdot \frac{m_3}{V_3}(F_3^{out} + D_3)\zeta_3 - \frac{1}{R_3} \cdot \frac{m_2}{V_2}F_2^{out}\zeta_2\frac{A_2}{A_3} \tag{26}$$

Ontario

$$c_{r,4} = \frac{1}{R_4} \cdot \frac{m_4}{V_4}F_4^{out}\zeta_4 - \frac{1}{R_4} \cdot \frac{m_3}{V_3}(F_3^{out} + D_3)\zeta_3\frac{A_3}{A_4} \tag{27}$$

Implement our model:

If we want to simulate the propagation of pollution in the Great Lakes when there were 20,000 t of pollution in Lake Superior in 1950. Assuming each lake has the same value of c_r, which is $2.6 \times 10^{-5}\ kg/m^3$.

Then substitute the required parameters into our code of the model:

```
%Initial Values
%kg
m1_1(1) =100000 * 1E3 ;
m2_1(1) = 0;
m3_1(1) = 0;
m4_1(1) = 0;
```

```
% concentration of pollution (kg/m3)
cir=2.6 * 1E-5;
rho1_1 = zeros(size(dV1));
rho2_1 = zeros(size(dV1));
rho3_1 = zeros(size(dV1));
rho4_1 = zeros(size(dV1));
```

The result run by our model is:

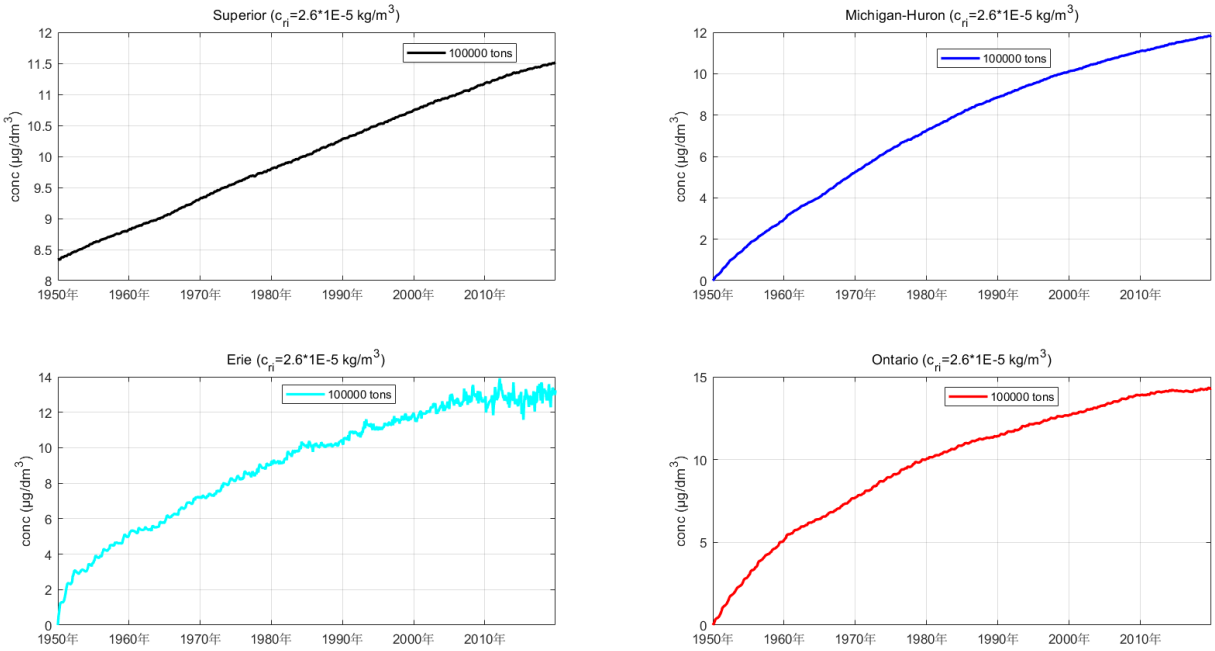

Figure 2: Stimulated propagation of pollution in the Great Lakes

The full version of the MATLAB code can be seen in the Appendix

Results and Discussion

Research Question 1

Diffusion is the process done by many things in the world, like atoms, molecules, particles, etc., which move from a region where they are highly concentration to a region, they are less concentrated. A concentration gradient is needed for them to move. This occurs in perfume being spread in the air, movement of electrons in pn junctions, etc. In our case, polluting material in lakes being located in one region of a large lake which connects to other lakes may from a highly concentrated area to a less concentrated area, thus spreading pollution to other lakes and to the regions of the same lake. The predominantly polluted lake of all Great Lakes was Lake Erie in the 1900s due to industries being located at its shores where sewage was dumped and toxic material.[9] Lake Superior is the cleanest lakes but has been still threatened by toxic pollutants and because of its long retention time outlet, 191 years [7], the pollutants remain in the lake for a very long time before being moved to other connecting lakes, Lake Michigan-Huron. We have created a model to describe the propagation of pollution through the Great Lakes considering the inflows and outflows of the lakes and the average approximate concentration of pollution in water runoff in the lakes. If we assume that the concentration of pollutants in water runoff is zero and that Lake Superior was polluted by approximately 75,000, 50,000, and 25,000 metric tons of pollution in 1950, we can present the propagation of those pollutants in the lakes through the model previously created. Minor changes need to be made to make this possible. We approximated the concentration of pollutants in water runoff, created by the heavy use of fertilizers in nearby agricultural landscape, to be about

$0.076 * 10^{-9} kgm^3$ and that the initial mass of pollutants in each lake in 1950 was 0 kg. However, we have assumed that the concentration of pollutants in water runoff was zero and that Lake Superior has initially 75,000, 50,000 or 25,000 metric tons of pollution stored in it in 1950. Therefore, we have to change the values and we can present the propagation of pollution in all of the lakes.

By doing so, we can observe the propagation of pollution downstream in the interconnecting lakes when the most upstream lake has been heavily polluted. This way we are able to prove which lake of all the Great Lakes has the highest risk of getting heavily polluted, therefore, we are able to provide more treatment to the more heavily polluted lakes in order to sustain a healthy ecosystem in the lakes which provide as a good food supply, industrial cooling system and recreational pursuits. For example, Lake Erie is the warmest of the lakes, the shallowest of the lakes and has the most sewage and agricultural runoff deposited in it. [1]Therefore, providing a great ecosystem for algae blooms to form, since it's the warmest, and toxic materials affect it the most, since it is the shallowest.

The following figure represents the propagation concentration of pollution, in $\mu g/dm^3$ where $dm^3 = litre$, in each lake from 1950 up until 2019 when Lake Superior was polluted by 75,000 metric tons of pollution.

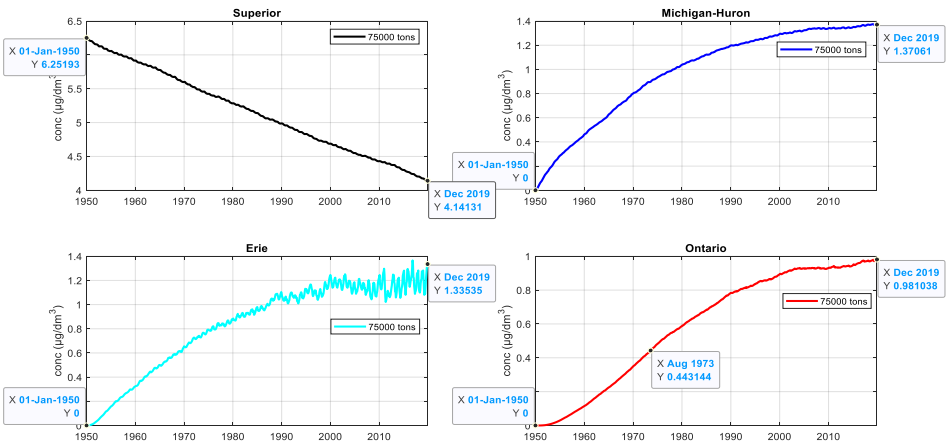

Figure 2: Graphical representation of the propagation of concentration of pollution in the Great Lakes, conc. of pollutants (µg/dm^3) against time (years).

In Fig. (2) it is obvious that Lake Superior has had its concentration of pollutants decrease in approximately at a constant rate from $6.252\ \mu g/dm^3$ in Jan 1950 to $4.141\ \mu g/dm^3$ in Dec 2019. Lake Michigan-Huron is connected to Lake Superior and the pollutants start to diffuse to Lake Michigan-Huron due to the concentration gradient. As assumed, the initial concentration of pollutants in the other three Great Lakes are $0\ \mu g/dm^3$ in 1950. The concentration of pollutants in lake Michigan-Huron increased at a decreasing rate throughout the 70 years up to $1.371\ \mu g/dm^3$. The next downstream lake is Lake Erie. The concentration of pollutants in Lake Erie started increasing at a decreasing rate, however, as time went by many fluctuations, we visible, shown in Fig. (2). This may be due to many factors such as seasons, during winter there is more precipitation and less evaporation but in summer precipitation

[1] https://www.youtube.com/watch?v=gMwQaHtK904&ab_channel=KQEDQUEST

decreases and evaporation decreases. Additionally, the changes from day and night may cause fluctuations due to the change in temperature and change in water flow in connecting channels of the lakes caused by the relative distance between the Earth and the Moon during their orbits. However, these factors would most likely affect the other three lakes as well. Lake Erie is the most biologically productive lake due to its warm shallow waters. It is also used for drinking water and located near its shore are many agricultural lands.[10] A lot of the water runoff from the agricultural lands end up in Lake Erie. This may be the leading cause of the fluctuations since specific fruits and vegetables require a specific season to be grown.

Lastly the concentration of pollutants in Lake Ontario increases at an increasing rate until 1973 to 0.443 $\mu g/dm^3$ and then the concentration of pollutants began to increase at a decreasing rate to $0.981\ \mu g/dm^3$. The lake with the highest concentration of pollutants was Lake Superior, this could be because the Lake Superior has a very long retention time outlet of 191 years, therefore, not all of the pollutants managed to leave it, due to its large volume.

he following figure represents the propagation of pollution from 1950 to 2019 when Lake Superior was polluted by 50,000 metric tons of pollution.

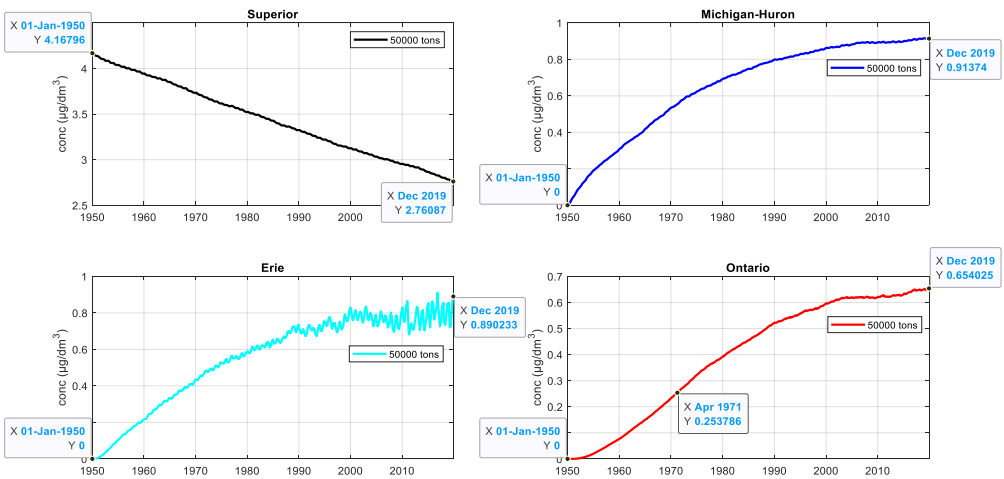

Figure 3: Graphical representation of the propagation of concentration of pollution in the Great Lakes, conc. of pollutants (µg/dm^3) against time (years).

In Fig. (3) Lake Superior initially has a lower concentration of pollutants than in Fig. (2). It has a concentration of $4.168\ \mu g/dm^3$ in 1950. The concentration of pollutants decreased at approximately a constant rate to $2.761\ \mu g/dm^3$ in 2019. The concentration of pollutants in lake Michigan-Huron increased at a decreasing rate to $0.931\ \mu g/dm^3$. Lake Erie had fluctuations in concentration at the same dates, however, at lower amplitudes. The concentration of pollutants firstly increased at a decreasing rate, then increased with fluctuations up to $1.335\ \mu g/dm^3$ in 2019. Lastly, Lake Ontario increased again at an increasing rate to $0.254\ \mu g/dm^3$ until 1971. It then began increasing at a decreasing rate to $0.654\ \mu g/dm^3$ in 2019. Ultimately, the concentration of pollutants in each lake decreased with Lake Ontario having the least concentration of pollutants since the data was only recorded for 70 years.

Finally, Lake Superior is polluted with 25,000 metric tons of pollution, the following graph describes its propagation.

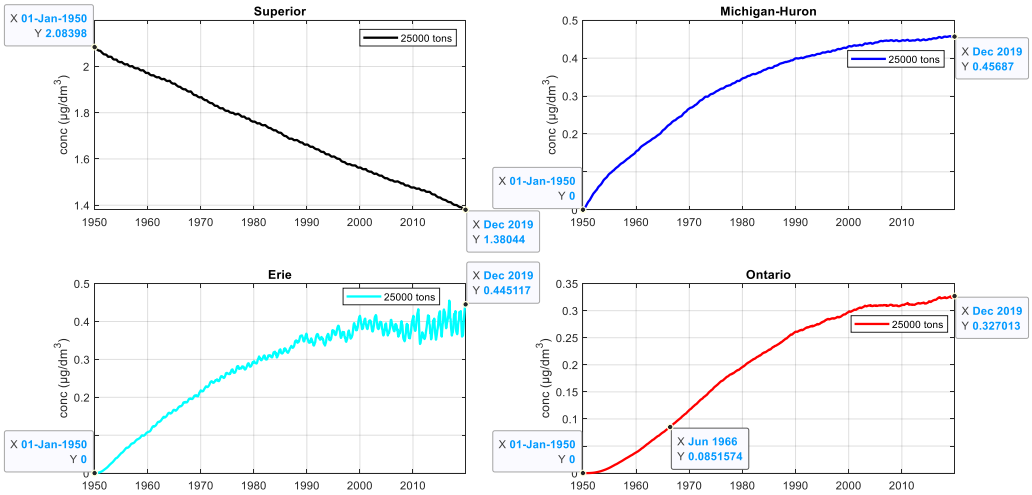

Figure 4: Graphical representation of the propagation of concentration of pollution in the Great Lakes, conc. of pollutants (µg/dm^3) against time (years).

In Fig. (4) Lake Superior has an initial concentration of pollutants of $2.083\ \mu g/dm^3$ in 1950. It decreased at a constant rate to $1.380\ \mu g/dm^3$ in 2019. Lake Michigan-Huron increased at a decreasing rate to $0.457\ \mu g/dm^3$ in 2019. Lake Erie increased at a decreasing rate until it began fluctuating. In 2019 the concentration of pollutants was about $0.445\ \mu g/dm^3$, this will probably continue fluctuating. Lastly, Lake Ontario increased at an increasing rate up to 1956 where the concentration of pollutants was $0.085\ \mu g/dm^3$. The concentration of pollutants then increased at a decreasing rate up to $0.327\ \mu g/dm^3$ in 2019.

In conclusion, as the mass of pollutants initially located in Lake Superior decreased the concentration of pollutants throughout the 70 years would decrease. However, the propagation of the concentration of pollution will be approximately the same for each lake. If the recording time was longer we would probably be able to see the concentration of pollutants in each lake ultimately be approximately the same because of diffusion.

Research Question 2

In Research Question 2, the concentration of pollutants in the water runoff, $c_{r,i}$ are no longer assumed as zero. Aiming to observe how the value of $c_{r,i}$ affect the distribution of pollutants and simplify the analysis. It has been assumed that the values of $c_{r,i}$ are the same for each lake, and also assumed each lake is initially clean in 1950. i.e., $m_i = 0\ kg$ in 1950, $i \in \{1,2,3,4\}$.

The results of Question 2 ($c_{r,i} = 2.601 \times 10^{-5}\ kg/m^3$):

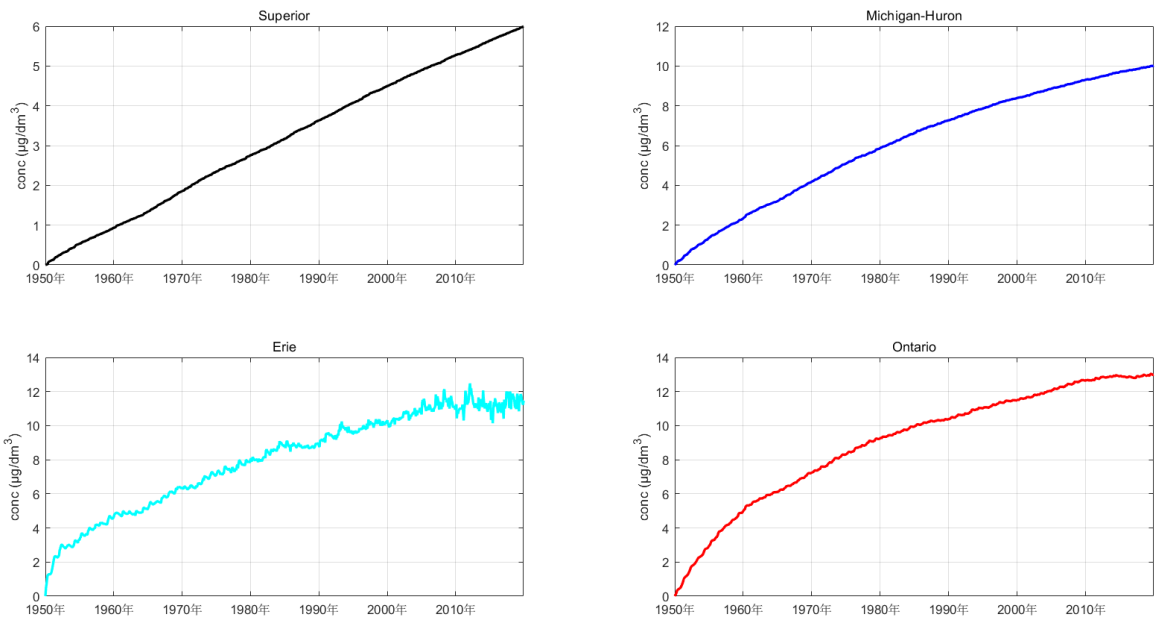

Figure 5: propagation of pollution in the Great Lakes when$c_{r,i} = 2.601 \times 10^{-5}\ kg/m^3$

Trends comparison

Figure.1 clearly shows that the pollutant concentration in each lake mainly had an increasing trend. The increasing rate of pollutant concentration in Lake Superior was approximately constant, while in the other lakes, the increasing rates were getting slower from 1950 to 2019.

By comparing the magnitude of the concentration of pollutant in each lake, pollutant was relatively more concentrated in downstream lakes, especially for Lake Erie and Ontario, whose values were over twice that of the Lake Superior

Estimating the value of $c_{r,i}$:

It is not realistic to give a definite value of $c_{r,i}$ at the current stage, since its value can be affected by various factors. To simplify the analyses, we have some assumptions

to estimate the $c_{r,i}$. We first assumed the main pollutant in the Great Lakes is phosphor.

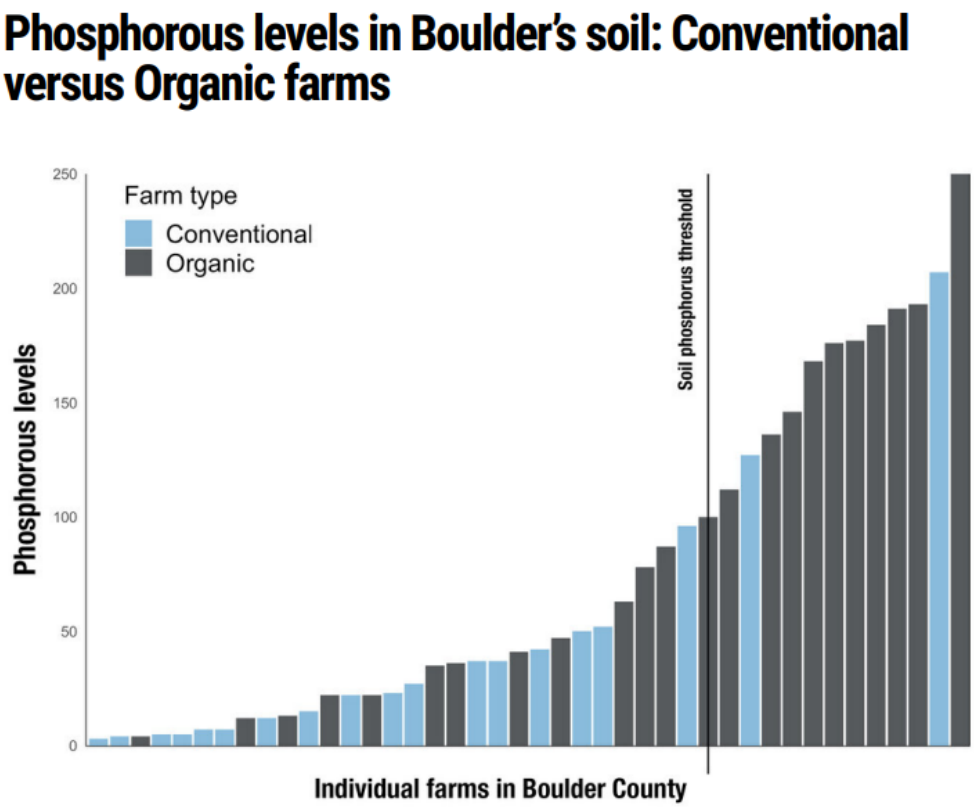

Figure 6: Phosphorus levels (mg P kg-1) in the soil of conventional versus organic farms in Boulder County.[8]

The average phosphorus levels for all of these farms (including conventional and organic) is around $75\ mg/kg^{-1}$ of soil.

These soils are likely to be picked up by water from rainfall which eventually results in water runoff in the Great Lakes. Assuming the bulk density of the soil is $1520\ kg/m^3$, and the ratio of soil and water in $1\ m^3$ of muddy water (mixture of water and soil) is:

$$\frac{soil}{clean\ water} = \frac{3}{7}$$

Then the concentration of the muddy water can be calculated:

$$\rho_{muddy} = \rho_{soil} \times 0.3 + \rho_{water} \times 0.7 = 1520 \times 0.3 + 1000 \times 0.7 = 1156\ kg/m^3 \qquad (28)$$

Since the concentrations of dissolved phosphorus in water running through fertilised soil are much smaller than the concentration of phosphorus in the soil,[6] which means the water from rainfall cannot fresh away all of the phosphor in the soil, but

usually greater than $0.02\ mg/L$.[5] Therefore we assumed that around 1/1000 of the phosphor per kg of the soil will dissolve in the rainfall water. So, the phosphor level in the soil in the water off is:

$$\text{Soil phosphor level in runoff} = \frac{\text{Average soil phosphor level on land}}{1000} = 75\ \mu g/kg^{-1}$$

Thus, the concentration of phosphors in the water runoff, $c_{r,i}$ can be calculated:

$$c_{r,i} = 1156\frac{kg}{m^3} \times 0.3 \times 75\frac{\mu g}{kg^{-1}} = 2.601 \times 10^{-5}\ kg/m^3 \tag{29}$$

Investigating the relative importance of the mechanisms of pollution

The main mechanisms of pollution are the flow in from the upstream lake, the flow out to the downstream lake, and the water runoff. In order to find which is the main factor that defines the concentration of pollution, comparing the flow rate of each factor can be helpful, which is shown in Fig. (7)

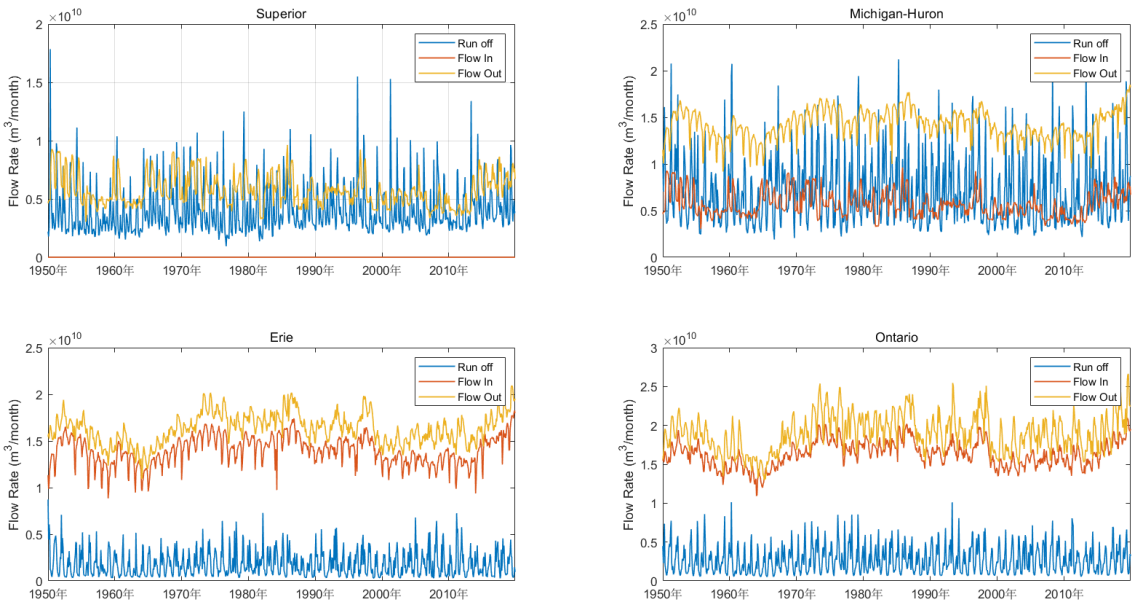

Figure.7 flow rates comparison of the mechanisms of pollution

Figure.5 has shown that downstream lakes are relatively more heavily polluted than upstream lakes. By looking at Figure.3, the flow rates of these lakes are all around $0{\sim}1\ m^3/month$, therefore water runoff cannot be the defining factor of the pollutant concentration.

Although for each lake, the outflow rate is always above its inflow rate, the difference between these two flow rates is decreasing from Lake Superior to Lake Ontario. According to Figure.5 and 7, we have also found that the smaller this difference in a

lake, the greater the pollutant concentration will be in that lake. Thus, the difference in inflow and outflow rates should be the main reason affecting the concentration of pollution.

According to Equation (8) – (15), the inflow of one lake depends on the outflow from its upstream lake:

Superior $$T_1^{in} = 0, \qquad T_1^{out} = F_1^{out} \tag{30}$$

Michigan-Huron $$T_2^{in} = F_1^{out}, \qquad T_2^{out} = F_2^{out} + D_2 \tag{31}$$

Erie $$T_3^{in} = F_2^{out}, \qquad T_3^{out} = F_3^{out} + D_3 \tag{32}$$

Ontario $$T_4^{in} = F_3^{out} + D_3, \qquad T_4^{out} = F_4^{out} \tag{33}$$

So,

$$T_1^{in} = 0, \qquad T_2^{in} = T_1^{out}, \qquad T_3^{in} \approx T_2^{out}\ (shown\ in\ Figure.7), \qquad T_4^{in} = T_3^{out} \tag{34}$$

However, the outflow from an upstream lake is not directly related to the outflow from the downstream lake. e.g. $T_1^{out} is\ not\ directly\ related\ T_2^{out}$. Instead, the outflow rate of a lake is related to some physical properties of the lake itself. e.g., the lake level and the capacity of the effluent channel. Therefore, the difference between the inflow and outflow rates of a lake depends on the outflow of the lake. If we assume that the pollutant concertation in inflow water and outflow water is the same, the self-clean ability of the lakes gets weaker from the upstream to the downstream. Consequently, the pollutants are likely to accumulate in downstream lakes.

Therefore, the outflow rate may be the defining factor of the concertation of pollution.

Research Question 3

Using the value of $c_{r,i}$ estimated in Research Question 2, which is $2.60 \times 10^{-5}\ kg/m^3$. According to Fig. (3), the concentration of pollution (phosphorus) in Lakes Erie and Ontario is between 10 and 20 $\mu g/L$ which enables normal growth of algae (mesotrophic). By increasing the value of $c_{r,i}$ gradually, we find the value of $c_{r,i}$ is above $4.6 \times 10^{-5}\ kg/m^3$, the pollutant concentrations in those lakes become just above $20\ \mu g/L$, which can result in that can cause a eutrophic state of the water.

Phosphorus is commonly used as a fertilizer in agriculture, and excessive use or poor management of phosphorus-containing fertilizers can lead to phosphorus runoff into nearby lakes and rivers. To verify the effectiveness of reducing eutrophication by imposing the recommended threshold concentration of phosphorus per kilogram of soil. We take the average phosphorus levels for the farms that surpass the threshold and those below in Fig. (4), which are $172\ mg\ kg^{-1}$ and $30\ mg\ kg^{-1}$. By following the method in Research Question 2 to find $c_{r,i}$, the estimated $c_{r,i}$'s are $5.96 \times 10^{-5}\ kg/m^3$ and $1.04 \times 10^{-5}\ kg/m^3$ respectively.

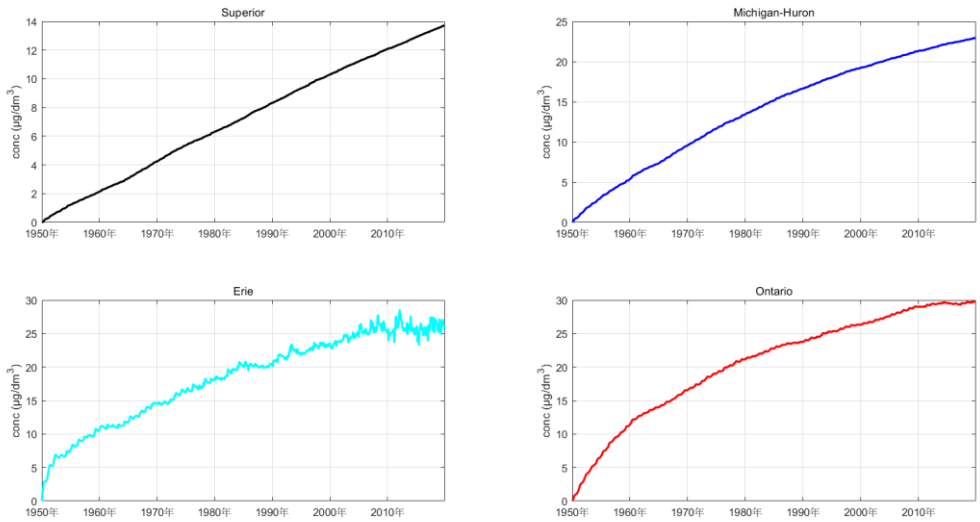

Figure.8 propagation of pollution in the Great Lakes when $c_{r,i} = 5.96 \times 10^{-5}\ kg/m^3$

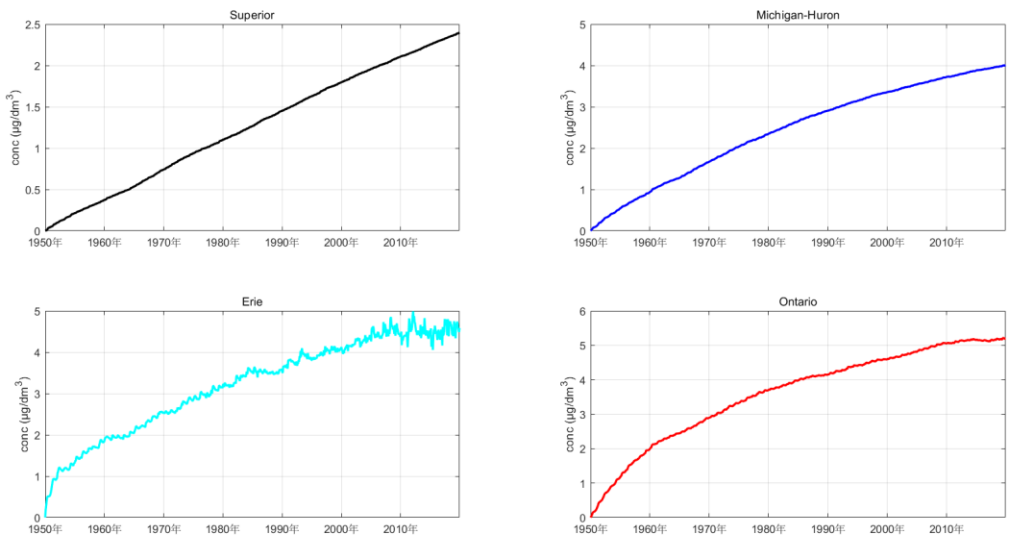

Figure.9 propagation of pollution in the Great Lakes when $c_{r,i} = 1.04 \times 10^{-5}\ kg/m^3$

By comparing Fig. (8) and (9), the change in the value of $c_{r,i}$ can significantly affect the pollutant concentration in the Great lakes. Thus, limiting the recommended threshold concentration of phosphorus per kilogram of soil could be an effective policy for reducing eutrophication and agricultural activities could be one of the major sources of phosphorus runoff into the lakes.

However, there are also ethical issues to consider. Limiting the amount of phosphorous applied per unit of soil could increase the cost of production for farmers and may considerably affect small or low-income farmers who can't afford to adopt alternative management practices. Besides, only 10.5% of conventional farms, compared to 47.8% of organic farms, are at or above the recommended phosphorous threshold, which means organic farmers will afford the main economic loss of limiting phosphorus policy. [5]

Additionally, limiting phosphorus use may lead to decreased crop yields, especially organic crops, which could have negative impacts on food security, and global food supply, and people with a specific demand for healthier food must spend more on the consumption of organic products.

Therefore, this policy aimed at limiting phosphorus use in agriculture should consider these ethical issues and try to mitigate any potential negative impacts on farmers and food security. Government should provide financial incentives or technical support to farmers who adopt alternative management practices.

However, in addition to the phosphorus in soil because of agricultural activities, many other factors lead to the five lakes being polluted. The first is Industrial Pollution. The Great Lakes are traditional industrial areas, including factories and power station, which produces many pollutants including heavy metals and harmful chemicals. Also, the development of industry in these areas attracted many populations, which lead to the amount of sewage that enters the lakes, carrying bacteria and pollutants. For example, much phosphorous in detergent is also carried by sewage into lakes, which also increases the amount of phosphorous.

Limitations

Furthermore, it is important to acknowledge that the research may not have a definitive or seemingly correct answer. In addition to limitations of research 3 that has been discussed in following questions. We must explain the limitations of the study and provide an explanation in research 1, and 2.

Firstly, in research 1, the inflow polluted from runoff is assumed to be 0, which means we only think about how pollution in Lake Superior propagates in all of the lakes, ideally, and it is impossible that pollution in Superior decreases at constant rate because inflow polluted from runoff increase the pollution in Superior.

In research 2, it is reasonable that the inflow polluted from runoff is no longer as 0, and it is estimated to be $c_{r,i} = 2.601 \times 10^{-5}\ kg/m^3$ by thinking about the average phosphorus in farms and how much phosphorus dissolved in water runoff after precipitation. But process of estimation is still idealistic because water in different temperature has different solubility, which means there is less amount of phosphorus in water runoff in winter than in summer. Also, speed of runoff can affect the amount of phosphorus. For example, water runoff in higher slope flow much more rapidly than in flat, which causes more soil containing phosphorus that is flush into lakes.

Conclusion

In research 1, $c_{r,i}$ is assumed to be 0 and Superior was polluted by approximately 75,000, 50,000, and 25,000 metric tons of pollution in 1950. We found how the pollution propagate in lakes of superior, Michigan-Huron, Erie and Ontario in Fig. (2), 5, 6, and whatever the value of $c_{r,i}$ is, the trend of propagation of pollution is similar from 1950 to 2019. The decreasing rate of pollution in superior is basically constant, because we assumed there is no new pollution that can flow into the superior, but the pollution of superior is still highest in 2019 due to a very long retention time outlet of 191 years. For lakes of Michigan-Huron, Erie and Ontario, the pollution concentration increases rapidly at first and tends to be constant finally. But the final stage of Erie is more and more fluctuated around a constant average value. Additionally, as more metric tons of pollution put into superior in 1950, the concentration of pollution would increase throughout 70 years in despite of same propagation trends.

In research 2, $c_{r,i}$ are no longer assumed as zero, and we must think about the concentration of pollutants in the water runoff, From the data from Fig. (6), we have found the value of $c_{r,i}$ by thinking about the average phosphorus in farms and how much phosphorus dissolved in water runoff after precipitation.

$$c_{r,i} = 2.601 \times 10^{-5}\ kg/m^3.$$

The diagram demonstrates the increasing rate of pollutant concentration in Lake Superior was approximately constant, but the increasing rates were getting slower

from 1950 to 2019, and the pollutant was relatively more concentrated in these downstream lakes

By looking at Fig. (5), the flow rates of these lakes are all around $0{\sim}1\ m^3/month$, which means it doesn't affect the value of pollutant concentration. However, from Fig. (5), we found if the value of "Flow in" and "Flow out" are more similar, the pollutant concentration of the lake is higher.

In research 3, we found the value of $c_{r,i}$ is above $4.6 \times 10^{-5}\ kg/m^3$ when the pollutant concentrations in Lakes Erie and Ontario become above $20\ \mu g/L$ (eutrophic state of the water). Then we set 2 different $c_{r,i}$, and compare the Fig. (8) and (9). We found the higher $c_{r,i}$ above the threshold causes much more pollutant concentration in great lakes. Thus, the limiting concentration of phosphorus could be an effective policy.

However, this policy gives rise to some ethical issues. Decreasing amount of Phosphorus in soil is negative to food security, because phosphorus is essential for growth of crops, the limiting policy could decrease the crops yield, especially organic crops which grow in the high phosphorus soil. In addition, farmers must adopt the alternative management without too much phosphorus, which perhaps is not affordable for small or low-income farmers and the organic farmers because 47.8% of organic farms are above the recommended phosphorous threshold.

Besides, we have some limitations in our model. As traditional industrial areas, we didn't think about many pollutants including heavy metals and harmful chemicals, produced by many factories, as well as the phosphorous in detergent carried by sewage into lakes in addition to the phosphorous from farms.

Reference

[1] Beeton, A.M. (2018). Great Lakes | Names, Map, & Facts. In: *Encyclopædia Britannica*. [online] Available at: https://www.britannica.com/place/Great-Lakes.

[2] US EPA, O. (2015). *Agriculture Nutrient Management and Fertilizer*. [online] US EPA. Available at: https://www.epa.gov/agriculture/agriculture-nutrient-management-and-fertilizer#:~:text=Most%20fertilizers%20that%20are%20commonly.

[3] US Department of Commerce, N.O. and A.A. (2022). *What is eutrophication?* [online] oceanservice.noaa.gov. Available at: https://oceanservice.noaa.gov/facts/eutrophication.html#:~:text=the%20nation.

[4] US EPA,REG 05 (2015). *Recommended Binational Phosphorus Targets | US EPA*. [online] US EPA. Available at: https://www.epa.gov/glwqa/recommended-binational-phosphorus-targets.

[5] Hussain, M.Z., Hamilton, S.K., Robertson, G.P. and Basso, B. (2021). Phosphorus availability and leaching losses in annual and perennial cropping systems in an upper US Midwest landscape. *Scientific Reports*, [online] 11(1), p.20367. doi: https://doi.org/10.1038/s41598-021-99877-7.

[6] Alabama Cooperative Extension System. (n.d.). *Phosphorus Basics: Understanding Pathways of Soil Phosphorus Loss*. [online] Available at: https://www.aces.edu/blog/topics/crop-production/understanding-soil-phosphorus-loss/#:~:text=Loss%20of%20phosphorus%20from%20land [Accessed 9 Mar. 2023].

[7] coastwatch.glerl.noaa.gov. (n.d.). *Great Lakes Physical Characteristics*. [online] Available at: https://coastwatch.glerl.noaa.gov/statistic/physical.html.

[8] 2020 Trends Report. Center for Sustainable Landscapes and Communities. Available at: https://cslc.colorado.edu/2020-trends/organic-farms-have-higher-phosphorous-levels

[9] Umich.edu. (2017). *Great Lakes Pollution · Exhibit · Give Earth a Chance: Environmental Activism in Michigan*. [online] Available at: https://michiganintheworld.history.lsa.umich.edu/environmentalism/exhibits/show/main_exhibit/pollution_politics/great-lakes-pollution#:~:text=Of%20all%20of%20the%20Great.

[10] Guide, G.L. (n.d.). *Lake Erie*. [online] Great Lakes Guide. Available at: https://greatlakes.guide/watersheds/erie#:~:text=Approximately%2033%25%20of%20the%20Great [Accessed 10 Mar. 2023].

Appendix

Model 1 (used to plot Figure.2, 3, 4, 5):

```
% SECTION 1 - Load variables
clear
close all
clc
% Replace the path within quotation marks below to the folder where the
% data is stored in your computer
cd 'C:\Users\ROG\Desktop\Data\Flow Data'
FL = dir('*.csv');

% Read the spreadsheet with dates and transform it into an array
time = readtable('Date.xlsx');
time = table2array(time);

% surface areas and volumes
A1 = 8.1925e10;                     % Superior
V01 = 12000*1E9;
A2 = 1.1685e11;                     % Mi-Huron
V02 = (3500+4900)*1E9;
A3 = 2.5404e10;                     % Erie
V03 = 480 * 1E9;
A4 = 1.9121e10;                     % Ontario
V04 = 1640 * 1E9;

% load diversions
% Terms multiplied convert data from m3/s to mm over the lakes surface area
D1 = readmatrix('D1.csv')*1000*60*60*24*30/A1;
D2 = readmatrix('D2.csv')*1000*60*60*24*30/A2;
D3 = readmatrix('D3.csv')*1000*60*60*24*30/A3;

% load connecting flows
F1 = readmatrix('Fo1.csv')*1000*60*60*24*30/A1;
F2 = readmatrix('Fo2.csv')*1000*60*60*24*30/A2;
F3 = readmatrix('Fo3.csv')*1000*60*60*24*30/A3;
F4 = readmatrix('Fo4.csv')*1000*60*60*24*30/A4;

% load evaporation
E1 = readmatrix('E1.csv');
E2 = readmatrix('E2.csv');
E3 = readmatrix('E3.csv');
E4 = readmatrix('E4.csv');

% load precipitation
P1 = readmatrix('P1.csv');
P2 = readmatrix('P2.csv');
P3 = readmatrix('P3.csv');
P4 = readmatrix('P4.csv');

% load runoff
R1 = readmatrix('R1.csv');
R2 = readmatrix('R2.csv');
R3 = readmatrix('R3.csv');
R4 = readmatrix('R4.csv');

% compute ODEs 5.1 to 5.4
% Multiplication by A_i and 1e-3 converts to m3 per month
```

```
dV1 = (  R1 + P1 + D1 - F1 - E1) * A1                        * 1E-3;
dV2 = ( (R2 + P2 - D2 - F2 - E2) * A2 +       (F1 * A1) )   * 1E-3;
dV3 = ( (R3 + P3 - D3 - F3 - E3) * A3 +       (F2 * A2) )   * 1E-3;
dV4 = ( (R4 + P4      - F4 - E4) * A4 + (D3 + F3) * A3  )   * 1E-3;

% Integration loops. Timestep = 1month
% Notice that V_i (in this case) is simply the cumulative sum of the
% differentials. For concentration solutions a Euler scheme needs to be
% implemented
V1 = zeros(size(dV1));
V2 = zeros(size(dV2));
V3 = zeros(size(dV3));
V4 = zeros(size(dV4));

for in = 1 : length(dV1)
    V1(in) = sum(dV1(1:in));
    V2(in) = sum(dV2(1:in));
    V3(in) = sum(dV3(1:in));
    V4(in) = sum(dV4(1:in));
end

clear in
V1 = V1 + V01;
V2 = V2 + V02;
V3 = V3 + V03;
V4 = V4 + V04;


%% Code Implementation

%Initial Values (kg)

m1_1(1) = 100000 * 1E3 ;
m2_1(1) = 0;
m3_1(1) = 0;
m4_1(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr=2.6 * 1E-5;
rho1_1 = zeros(size(dV1));
rho2_1 = zeros(size(dV1));
rho3_1 = zeros(size(dV1));
rho4_1 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1_1(t) = m1_1(t) / V1(t);
    rho2_1(t) = m2_1(t) / V2(t);
    rho3_1(t) = m3_1(t) / V3(t);
    rho4_1(t) = m4_1(t) / V4(t);

    dm1_1 = (cr * R1(t)*A1 - rho1_1(t) * F1(t)*A1)* 1E-3;
    dm2_1 = (cr * R2(t)*A2 + rho1_1(t) * F1(t)*A1 - rho2_1(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3_1 = (cr * R3(t)*A3 + rho2_1(t) * F2(t)*A2 - rho3_1(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4_1 = (cr * R4(t)*A4 + rho3_1(t) * (F3(t) + D3(t))*A3 - rho4_1(t) *
F4(t)*A4)* 1E-3;

    m1_1(t+1) = m1_1(t) + dm1_1;
```

```
    m2_1(t+1) = m2_1(t) + dm2_1;
    m3_1(t+1) = m3_1(t) + dm3_1;
    m4_1(t+1) = m4_1(t) + dm4_1;

end

rho1_1(length(dV1)) = m1_1(length(dV1)) / V1(length(dV1));
rho2_1(length(dV1)) = m2_1(length(dV1)) / V2(length(dV1));
rho3_1(length(dV1)) = m3_1(length(dV1)) / V3(length(dV1));
rho4_1(length(dV1)) = m4_1(length(dV1)) / V4(length(dV1));


figure()
subplot(2,2,1)
plot(time, rho1_1 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Superior (c_r_i=2.6*1E-5 kg/m^3)')
legend("100000 tons")
grid on

subplot(2,2,2)
plot(time, rho2_1 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Michigan-Huron (c_r_i=2.6*1E-5 kg/m^3)')
legend("100000 tons")
grid on

subplot(2,2,3)
plot(time, rho3_1 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Erie (c_r_i=2.6*1E-5 kg/m^3)')
legend("100000 tons")
grid on

subplot(2,2,4)
plot(time, rho4_1 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Ontario (c_r_i=2.6*1E-5 kg/m^3)')
legend("100000 tons")
grid on

%% 75,000 metric tons of pollution in Lake Superior in 1950
m1 = zeros(size(dV1));
m2 = zeros(size(dV1));
m3 = zeros(size(dV1));
m4 = zeros(size(dV1));

%Initial Values (kg)

m1(1) = 75000 * 1E3 ;
m2(1) = 0;
m3(1) = 0;
m4(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr=0;
rho1 = zeros(size(dV1));
rho2 = zeros(size(dV1));
rho3 = zeros(size(dV1));
```

```
rho4 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1(t) = m1(t) / V1(t);
    rho2(t) = m2(t) / V2(t);
    rho3(t) = m3(t) / V3(t);
    rho4(t) = m4(t) / V4(t);

    dm1 = (cr * R1(t)*A1 - rho1(t) * F1(t)*A1)* 1E-3;
    dm2 = (cr * R2(t)*A2 + rho1(t) * F1(t)*A1 - rho2(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3 = (cr * R3(t)*A3 + rho2(t) * F2(t)*A2 - rho3(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4 = (cr * R4(t)*A4 + rho3(t) * (F3(t) + D3(t))*A3 - rho4(t) *
F4(t)*A4)* 1E-3;

    m1(t+1) = m1(t) + dm1;
    m2(t+1) = m2(t) + dm2;
    m3(t+1) = m3(t) + dm3;
    m4(t+1) = m4(t) + dm4;

end

rho1(length(dV1)) = m1(length(dV1)) / V1(length(dV1));
rho2(length(dV1)) = m2(length(dV1)) / V2(length(dV1));
rho3(length(dV1)) = m3(length(dV1)) / V3(length(dV1));
rho4(length(dV1)) = m4(length(dV1)) / V4(length(dV1));

figure()
subplot(2,2,1)
plot(time, rho1 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Superior')
legend("75000 tons")
grid on

subplot(2,2,2)
plot(time, rho2 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Michigan-Huron')
legend("75000 tons")
grid on

subplot(2,2,3)
plot(time, rho3 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Erie')
legend("75000 tons")
grid on

subplot(2,2,4)
plot(time, rho4 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Ontario')
legend("75000 tons")
grid on

%% 50,000 metric tons of pollution in Lake Superior in 1950
```

```
m1_2 = zeros(size(dV1));
m2_2 = zeros(size(dV1));
m3_2 = zeros(size(dV1));
m4_2 = zeros(size(dV1));

%Initial Values (kg)
m1_2(1) = 50000 * 1E3 ;
m2_2(1) = 0;
m3_2(1) = 0;
m4_2(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr = 0;
rho1_2 = zeros(size(dV1));
rho2_2 = zeros(size(dV1));
rho3_2 = zeros(size(dV1));
rho4_2 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1_2(t) = m1_2(t) / V1(t);
    rho2_2(t) = m2_2(t) / V2(t);
    rho3_2(t) = m3_2(t) / V3(t);
    rho4_2(t) = m4_2(t) / V4(t);

    dm1_2 = (cr * R1(t)*A1 - rho1_2(t) * F1(t)*A1)* 1E-3;
    dm2_2 = (cr * R2(t)*A2 + rho1_2(t) * F1(t)*A1 - rho2_2(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3_2 = (cr * R3(t)*A3 + rho2_2(t) * F2(t)*A2 - rho3_2(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4_2 = (cr * R4(t)*A4 + rho3_2(t) * (F3(t) + D3(t))*A3 - rho4_2(t) *
F4(t)*A4)* 1E-3;

    m1_2(t+1) = m1_2(t) + dm1_2;
    m2_2(t+1) = m2_2(t) + dm2_2;
    m3_2(t+1) = m3_2(t) + dm3_2;
    m4_2(t+1) = m4_2(t) + dm4_2;

end

rho1_2(length(dV1)) = m1_2(length(dV1)) / V1(length(dV1));
rho2_2(length(dV1)) = m2_2(length(dV1)) / V2(length(dV1));
rho3_2(length(dV1)) = m3_2(length(dV1)) / V3(length(dV1));
rho4_2(length(dV1)) = m4_2(length(dV1)) / V4(length(dV1));


figure()
subplot(2,2,1)
plot(time, rho1_2 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Superior')
legend("50000 tons")
grid on

subplot(2,2,2)
plot(time, rho2_2 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Michigan-Huron')
legend("50000 tons")
grid on
```

```
subplot(2,2,3)
plot(time, rho3_2 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Erie')
legend("50000 tons")
grid on

subplot(2,2,4)
plot(time, rho4_2 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Ontario')
legend("50000 tons")
grid on


%% 25,000 metric tons of pollution in Lake Superior in 1950

m1_3 = zeros(size(dV1));
m2_3 = zeros(size(dV1));
m3_3 = zeros(size(dV1));
m4_3 = zeros(size(dV1));

%Initial Values (kg)
m1_3(1) = 25000 * 1E3 ;
m2_3(1) = 0;
m3_3(1) = 0;
m4_3(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr = 0;
rho1_3 = zeros(size(dV1));
rho2_3 = zeros(size(dV1));
rho3_3 = zeros(size(dV1));
rho4_3 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1_3(t) = m1_3(t) / V1(t);
    rho2_3(t) = m2_3(t) / V2(t);
    rho3_3(t) = m3_3(t) / V3(t);
    rho4_3(t) = m4_3(t) / V4(t);

    dm1_3 = (cr * R1(t)*A1 - rho1_3(t) * F1(t)*A1)* 1E-3;
    dm2_3 = (cr * R2(t)*A2 + rho1_3(t) * F1(t)*A1 - rho2_3(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3_3 = (cr * R3(t)*A3 + rho2_3(t) * F2(t)*A2 - rho3_3(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4_3 = (cr * R4(t)*A4 + rho3_3(t) * (F3(t) + D3(t))*A3 - rho4_3(t) *
F4(t)*A4)* 1E-3;

    m1_3(t+1) = m1_3(t) + dm1_3;
    m2_3(t+1) = m2_3(t) + dm2_3;
    m3_3(t+1) = m3_3(t) + dm3_3;
    m4_3(t+1) = m4_3(t) + dm4_3;

end

rho1_3(length(dV1)) = m1_3(length(dV1)) / V1(length(dV1));
rho2_3(length(dV1)) = m2_3(length(dV1)) / V2(length(dV1));
```

```
rho3_3(length(dV1)) = m3_3(length(dV1)) / V3(length(dV1));
rho4_3(length(dV1)) = m4_3(length(dV1)) / V4(length(dV1));

figure()
subplot(2,2,1)
plot(time, rho1_3 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Superior')
legend("25000 tons")
grid on

subplot(2,2,2)
plot(time, rho2_3 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Michigan-Huron')
legend("25000 tons")
grid on

subplot(2,2,3)
plot(time, rho3_3 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Erie')
legend("25000 tons")
grid on

subplot(2,2,4)
plot(time, rho4_3 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
title('Ontario')
legend("25000 tons")
grid on


%% Research Question 2 (using estimated cri)
m1 = zeros(size(dV1));
m2 = zeros(size(dV1));
m3 = zeros(size(dV1));
m4 = zeros(size(dV1));

%Initial Values (kg)
m1(1) = 0;
m2(1) = 0;
m3(1) = 0;
m4(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr = 2.601 * 1E-5;
rho1 = zeros(size(dV1));
rho2 = zeros(size(dV1));
rho3 = zeros(size(dV1));
rho4 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1(t) = m1(t) / V1(t);
    rho2(t) = m2(t) / V2(t);
    rho3(t) = m3(t) / V3(t);
    rho4(t) = m4(t) / V4(t);

    dm1 = (cr * R1(t)*A1 - rho1(t) * F1(t)*A1)* 1E-3;
```

```
    dm2 = (cr * R2(t)*A2 + rho1(t) * F1(t)*A1 - rho2(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3 = (cr * R3(t)*A3 + rho2(t) * F2(t)*A2 - rho3(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4 = (cr * R4(t)*A4 + rho3(t) * (F3(t) + D3(t))*A3 - rho4(t) *
F4(t)*A4)* 1E-3;

    m1(t+1) = m1(t) + dm1;
    m2(t+1) = m2(t) + dm2;
    m3(t+1) = m3(t) + dm3;
    m4(t+1) = m4(t) + dm4;
end

rho1(length(dV1)) = m1(length(dV1)) / V1(length(dV1));
rho2(length(dV1)) = m2(length(dV1)) / V2(length(dV1));
rho3(length(dV1)) = m3(length(dV1)) / V3(length(dV1));
rho4(length(dV1)) = m4(length(dV1)) / V4(length(dV1));


figure()
subplot(2,2,1)
plot(time, rho1 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=2.601 * 1E-5 kg/m^3")
title('Superior')

grid on

subplot(2,2,2)
plot(time, rho2 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=2.601 * 1E-5 kg/m^3")
title('Michigan-Huron')

grid on

subplot(2,2,3)
plot(time, rho3 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=2.601 * 1E-5 kg/m^3")
title('Erie')

grid on

subplot(2,2,4)
plot(time, rho4 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=2.601 * 1E-5 kg/m^3")
title('Ontario')

grid on


%% Flow Rate Comparasion
figure()
subplot(2,2,1)
plot(time,R1*A1*1E-3,'-','linewidth',1)
hold on
y = zeros(size(dV1));
```

```
plot(time,y,'-','linewidth',1);
plot(time,F1*A1*1E-3,'-','linewidth',1)
ylabel("Flow Rate (m^3/month)")
legend("Run off","Flow In","Flow Out")
title('Superior')
hold off
grid on

subplot(2,2,2)
plot(time,R2*A2*1E-3,'-','linewidth',1)
hold on
plot(time,F1*A1*1E-3,'-','linewidth',1)
hold on
plot(time,(F2 + D2)*A2*1E-3,'-','linewidth',1)
ylabel("Flow Rate (m^3/month)")
legend("Run off","Flow In","Flow Out")
title('Michigan-Huron')
hold off

subplot(2,2,3)
plot(time,R3*A3*1E-3,'-','linewidth',1)
hold on
plot(time,F2*A2*1E-3,'-','linewidth',1)
hold on
plot(time,(F3 + D3)*A3*1E-3,'-','linewidth',1)
ylabel("Flow Rate (m^3/month)")
legend("Run off","Flow In","Flow Out")
title('Erie')
hold off

subplot(2,2,4)
plot(time,R4*A4*1E-3,'-','linewidth',1)
hold on
plot(time,(F3 + D3)*A3*1E-3,'-','linewidth',1)
hold on
plot(time,F4*A4*1E-3,'-','linewidth',1)
ylabel("Flow Rate (m^3/month)")
legend("Run off","Flow In","Flow Out")
title('Ontario')
hold off


%% Research Question 3
% cri=5.96 * 1E-5 kg/m^3
m1 = zeros(size(dV1));
m2 = zeros(size(dV1));
m3 = zeros(size(dV1));
m4 = zeros(size(dV1));

%Initial Values (kg)
m1(1) = 0;
m2(1) = 0;
m3(1) = 0;
m4(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr = 5.96 * 1E-5;
rho1 = zeros(size(dV1));
rho2 = zeros(size(dV1));
```

```
rho3 = zeros(size(dV1));
rho4 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1(t) = m1(t) / V1(t);
    rho2(t) = m2(t) / V2(t);
    rho3(t) = m3(t) / V3(t);
    rho4(t) = m4(t) / V4(t);

    dm1 = (cr * R1(t)*A1 - rho1(t) * F1(t)*A1)* 1E-3;
    dm2 = (cr * R2(t)*A2 + rho1(t) * F1(t)*A1 - rho2(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3 = (cr * R3(t)*A3 + rho2(t) * F2(t)*A2 - rho3(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4 = (cr * R4(t)*A4 + rho3(t) * (F3(t) + D3(t))*A3 - rho4(t) *
F4(t)*A4)* 1E-3;

    m1(t+1) = m1(t) + dm1;
    m2(t+1) = m2(t) + dm2;
    m3(t+1) = m3(t) + dm3;
    m4(t+1) = m4(t) + dm4;
end

rho1(length(dV1)) = m1(length(dV1)) / V1(length(dV1));
rho2(length(dV1)) = m2(length(dV1)) / V2(length(dV1));
rho3(length(dV1)) = m3(length(dV1)) / V3(length(dV1));
rho4(length(dV1)) = m4(length(dV1)) / V4(length(dV1));


figure()
subplot(2,2,1)
plot(time, rho1 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=5.96 * 1E-5 kg/m^3")
title('Superior')

grid on

subplot(2,2,2)
plot(time, rho2 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=5.96 * 1E-5 kg/m^3")
title('Michigan-Huron')

grid on

subplot(2,2,3)
plot(time, rho3 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=5.96 * 1E-5 kg/m^3")
title('Erie')

grid on

subplot(2,2,4)
plot(time, rho4 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=5.96 * 1E-5 kg/m^3")
title('Ontario')
```

```
grid on


% cri=1.04 * 1E-5 kg/m^3
m1 = zeros(size(dV1));
m2 = zeros(size(dV1));
m3 = zeros(size(dV1));
m4 = zeros(size(dV1));

%Initial Values (kg)
m1(1) = 0;
m2(1) = 0;
m3(1) = 0;
m4(1) = 0;

% Pollutant concretration and runoff concentration (kg/m3)
cr = 1.04 * 1E-5;
rho1 = zeros(size(dV1));
rho2 = zeros(size(dV1));
rho3 = zeros(size(dV1));
rho4 = zeros(size(dV1));

for t = 1 : (length(dV1) - 1)
    rho1(t) = m1(t) / V1(t);
    rho2(t) = m2(t) / V2(t);
    rho3(t) = m3(t) / V3(t);
    rho4(t) = m4(t) / V4(t);

    dm1 = (cr * R1(t)*A1 - rho1(t) * F1(t)*A1)* 1E-3;
    dm2 = (cr * R2(t)*A2 + rho1(t) * F1(t)*A1 - rho2(t) * (F2(t) +
D2(t))*A2)* 1E-3;
    dm3 = (cr * R3(t)*A3 + rho2(t) * F2(t)*A2 - rho3(t) * (F3(t) +
D3(t))*A3)* 1E-3;
    dm4 = (cr * R4(t)*A4 + rho3(t) * (F3(t) + D3(t))*A3 - rho4(t) *
F4(t)*A4)* 1E-3;

    m1(t+1) = m1(t) + dm1;
    m2(t+1) = m2(t) + dm2;
    m3(t+1) = m3(t) + dm3;
    m4(t+1) = m4(t) + dm4;
end

rho1(length(dV1)) = m1(length(dV1)) / V1(length(dV1));
rho2(length(dV1)) = m2(length(dV1)) / V2(length(dV1));
rho3(length(dV1)) = m3(length(dV1)) / V3(length(dV1));
rho4(length(dV1)) = m4(length(dV1)) / V4(length(dV1));


figure()
subplot(2,2,1)
plot(time, rho1 * 1E6, 'k', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=1.04 * 1E-5 kg/m^3")
title('Superior')

grid on

subplot(2,2,2)
```

```
plot(time, rho2 * 1E6, 'b', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=1.04 * 1E-5 kg/m^3")
title('Michigan-Huron')

grid on

subplot(2,2,3)
plot(time, rho3 * 1E6, 'c', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=1.04 * 1E-5 kg/m^3")
title('Erie')

grid on

subplot(2,2,4)
plot(time, rho4 * 1E6, 'r', 'linewidth', 2)
ylabel('conc (µg/dm^3)')
legend("c_r_i=1.04 * 1E-5 kg/m^3")
title('Ontario')

grid on
```